

Effects of Brightness on Distance Judgments in Head Mounted Displays

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ABSTRACT

Many HMD applications are improved when people perceive and act similarly in the virtual world as they would in an equivalent real environment. However, multiple studies show people underestimate egocentric distances in HMDs while being reasonably accurate in the real world. Some studies also show that brightness in the periphery may be one factor influencing these distance judgments. We built upon this previous work and examine if the brightness of the entire screen affects distance judgments. Using direct blind-walking, we compared a bright environment to an artificially darkened one with a between-subject experiment. The results of the experiment indicate that overall brightness has no significant effect on egocentric distance judgment. We discuss the implications of our findings in the context of previous work. Our findings may help designers of HMD software featuring dark environments (e.g., some entertainment applications).

CCS CONCEPTS

• **Human-centered computing** → *Virtual reality*; • **Computing methodologies** → *Perception*.

KEYWORDS

VR, HMD, distance judgments, perception, brightness

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1 INTRODUCTION AND BACKGROUND

VR provides an affordable and safe platform for a variety of applications including training, prototyping and entertainment. In many applications, it is crucial that people can understand simulated spaces as accurately as they can in corresponding real environments. The overall utility of VR is reduced if distance judgments are less accurate than the real world.

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There are numerous ways to measure absolute egocentric distance judgments (for a survey, see Renner et al. [2013]) in action space (2–30 meters). Since we cannot directly measure perceived distance, we must ask participants to verbally report the distance or perform an action that depends on the distance. For verbal reports, participants view a target and verbally indicate the distance in their preferred units. Andre and Rogers [2006] conclude that verbal reports are less accurate because they may be influenced by cognitive factors or by unfamiliarity with the units the participants use. Alternatively, the most common action-based measurement is direct blind-walking where participants walk blindfolded to a previously seen target and the distance walked is measured. In real environments where full cues are provided, people are accurate at distances up to 22 meters with direct blind-walking [Rieser et al. 1990]. Other less popular action-based techniques include triangulated walking, blind throwing [Sahm et al. 2005], timed imagined walking [Andre and Rogers 2006], or continuously pointing at a target while blind-walking [Fukushima et al. 2002].

While accurate in real environments, people often underestimate distances in HMDs [Renner et al. 2013]. Studies have tried to explain distance underestimation by examining a variety of factors including field of view [Buck et al. 2018; Jones et al. 2016; Knapp and Loomis 2004; Li et al. 2018, 2015], HMD mechanics [Willemssen et al. 2009], viewing condition [Willemssen et al. 2008], quality of graphics [Kunz et al. 2009; Thompson et al. 2004], and response measure [Grechkin et al. 2010; Sahm et al. 2005]. However, none have successfully identified the entire cause of distance underestimation.

Multiple studies have shown that manipulating geometric information can influence distance judgments. Notably, minification of the graphics presented to the user changes multiple distance cues and is known to influence distance judgments [Kuhl et al. 2009; Li et al. 2015; Zhang et al. 2012]. Geometric cues including the angle of declination to targets on the ground [Dilda et al. 2006; Wu et al. 2004] and familiar size [Kunz et al. 2009; Thompson et al. 2004] are thought to inform distance estimation. For targets beyond two meters, stereoscopic cues are unlikely to play a significant role [Willemssen et al. 2008].

It is not surprising that changes to the geometry presented to the user can change distance judgments. However, it may be surprising that changes to brightness can influence HMD distance perception. Some of the research on this topic focuses on brightness in the periphery. For example, Jones et al. [2011] found that adding LED “light bars” around HMD lenses reduced underestimation in distance judgments. Further, Li et al. [2018] used software to artificially reduce the rendered field of view and then varied the brightness in the peripheral area. The reduced field of view in this study was

matched an older HMD. With a dark periphery, normal amounts of distance compression were observed. However, another condition with a brighter peripheral treatment improved distance judgment accuracy. Despite this, there is no clear scientific explanation of why or how peripheral brightness influences distance judgments. Further, one might hypothesize that dim scenes in HMD environments may produce additional distance compression.

The earliest work on brightness and distance perception focused on reduced-cue environments. In these studies, people judged the absolute or relative distance of lit targets in darkened rooms and the target's brightness, hue, or saturation are varied [Coules 1955; Egusa 1982, 1983]. Since HMDs are typically used in full-cue, semi-realistic environments, more recent research focused on the influence of surface color is more relevant. For example, verbal reports suggest that bright rooms look larger than darker rooms in an HMD [Oberfeld et al. 2010; von Castell et al. 2018]. Reports of spaciousness also increased when a bright virtual room is displayed on a traditional monitor [Stamps III 2007]. However, limited research exists in full-cue real-world environments. For example, Matusiak [2004] viewers indicated on a linear scale that brighter scenes (with high illumination or brighter surfaces) tend to make spaces appear larger. We are unaware of previous work examining how overall scene brightness influences egocentric, absolute distance judgments with an action-based response measure.

Previous HMD studies that change the brightness in the periphery [Jones et al. 2012; Li et al. 2018], by definition, also change the overall average brightness presented to the viewer. Therefore, we cannot definitively distinguish between if peripheral brightness or overall average brightness is influencing distance judgments. Therefore, it is a natural to ask if changing the overall brightness of the scene might also have a similar impact on distance judgments. We feel that research into this area is important as it can inform the design of virtual environments (e.g., training simulations in dark environments) and HMD design. HMD manufacturers may be particularly interested in brightness since it impacts battery life on fully-portable HMDs and can be a factor related to eye strain [Chu et al. 2014].

2 EXPERIMENT

2.1 Design and Method

We rendered an environment in OpenGL on an Oculus CV1 HMD with two different brightnesses for our between-subject experiment (Figure 1). The virtual room was modeled after on campus lab and, textures were created from photographs. The virtual room contained some furniture around the perimeter and was also used by Li et al. [2018]. In the BRIGHT condition, the graphics were rendered as they normally would be in a typical HMD application. The 3D model included textures on all surfaces and was flatly shaded without specular or diffuse shading. The brightness of the room was determined by the brightness of the textures applied to the room and the brightness of the HMD itself. In the DARK condition, the graphics were rendered by reducing the relative luminance of the scene to 1% of the original brightness in an OpenGL fragment program. Assuming a typical gamma value of 2.2, this manipulation scaled the brightest value to 12% relative brightness in sRGB. In this condition, the scene is significantly darker but still bright enough

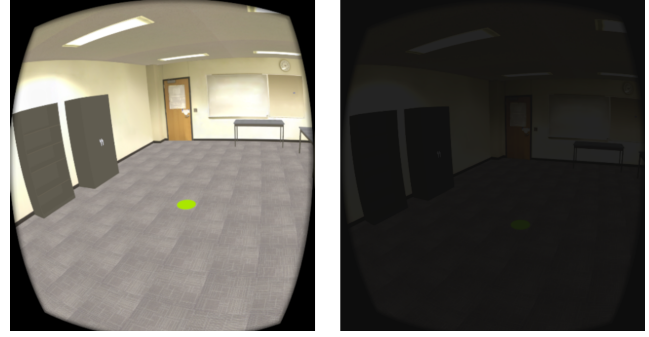


Figure 1: Left eye images for virtual environment in BRIGHT and DARK conditions with a target at three meters.



Figure 2: Virtual reality laboratory with Vicon tracking system.

that participants did not indicate that they had trouble viewing the scene. We found the DARK condition easier to view in the HMD than on the printed page because the dark scene is the brightest thing visible while wearing the HMD. Prior to viewing either condition, participants wore a blindfold for several minutes and their eyes were likely adapted to darkness.

The experiments took place in a 8×7 meter lab (Figure 2). The lab's 12-camera Vicon system tracked the HMD's position via infrared-reflective markers attached to the headset. Orientation was tracked using the Oculus's built-in sensor. However, to combat drifting, we also leveraged orientation data retrieved from the Vicon system. The tracking system was regularly calibrated to ensure reliable tracking, consistency, and coverage for all participants. There was sufficient space in the real world and virtual world for participants to walk normally without colliding with obstacles.

Our research and experiment procedure was followed the same as the procedure of similar paper by Li et al. [2018]. First, participants provided informed consent by completing an IRB form. We introduced the experiment to the participant and explained the procedure with written and verbal instructions. Participants were allowed to wear contacts or glasses if they fit inside the HMD. We verified that participants had corrected visual acuity of at least 20/30 with a Snellen eye chart. A random-dot stereogram test with stereoscope glasses ensured that participants were not stereoblind.

Participants were instructed not to use any counting strategies such as counting steps or counting floor tiles. They were also instructed not to perform the tasks as if they were math problems. Instead, participants were instructed to mentally visualize the scene that they were shown and to imagine how their viewpoint updated as they walked to the target with their eyes closed.

Since blind walking is an uncommon task that requires participants to trust the experimenter, we practiced blindfold walking with participants outside of the lab in the hallway with the same amount of distance and time to familiarize them with blind walking. The experimenter told participants to stop before running into any obstacle. Participants were instructed to turn before they started walking again. Next, they entered the lab blindfolded and we assisted them with putting on the HMD. Participants never saw the inside of the lab and had their eyes closed until the HMD was placed on them. White noise was played in the HMD headphones to mask real-world sounds. Masking these sounds prevented them from using acoustic cues during blind walking. The experimenter was always near the participant and spoke loudly so they could hear instructions over the white noise.

Next, each participant performed 17 direct blind-walking trials. For each trial, the participant was brought to the starting position and opened their eyes. They viewed the virtual environment and target for as long as they wished (typically around 15 seconds). They were allowed to rotate their heads, but instructed not to translate their body and head position. When our participants were ready to walk, they closed their eyes, said “OK”, the experimenter blanked the HMD screen, and the participant walked toward the target. They stopped when they felt that they were standing on top of the virtual target. We recorded their walked distance with the tracking system and guided them back to the starting position. The HMD screen was always blanked while the participant was walking. Consistent with most distance judgment studies, no feedback was given. If feedback had been provided after each trial, we would expect performance to improve due to adaptation.

The experiment began with two practice trials at distances of 3.5 and 4.5 in random order. Then, participants performed 15 experimental trials in random order. These trials had targets at distances of 2, 3, 4, and 5 meters. Each distance was shown three times. Three other trials at distances of 2.5, 3.5, and 4.5m (not repeated) were also included. These three trials were added to make it more difficult for participants to memorize all of the target distances we were displaying and they were not included in the analysis. In total, participants performed 17 blind walking trials. In each trial, we randomly selected the target from five shapes (circle, cross, triangle, etc.) ranging from 25-30cm wide. Each shape was also a different color. Several times throughout the trials, we verbally verified with participants that they were comfortable with completing the experiment.

The starting point, where participants start each trial and guided back to afterward, was fixed in the real world. However, the virtual room was randomly translated by 0, 0.5, and -0.5 meters along the walking path (i.e., forward or backward). This translation was *not* applied to the target and therefore had no impact on the target distance. For example, when a target distance is shown for a second time, the virtual world translational offset will often cause the target and starting position to appear at a different location relative to

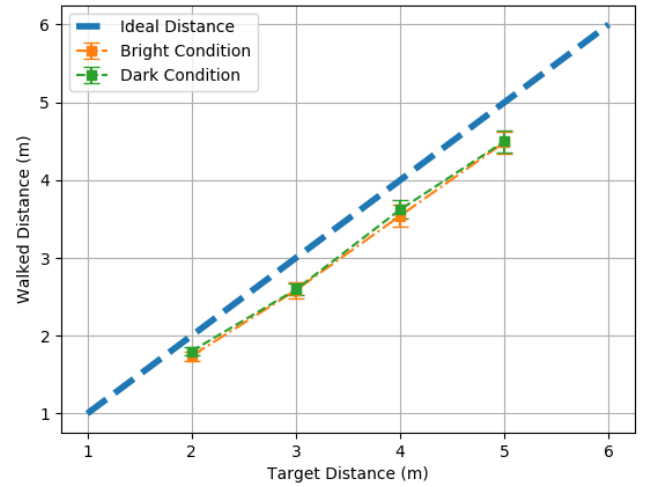


Figure 3: Results of distance judgments by direct blind-walking in the BRIGHT and DARK conditions. Error bars represent ± 1 SEM.

landmarks in the virtual room. We used this procedure to make it challenging for participants to recognize repeated target distances. After the experiment, participants completed a post-experiment questionnaire. It was designed to identify problems with the VR system by asking if participants noticed any errors in the displayed graphics and measure how many participants had previous experience with HMDs.

2.2 Results

The DARK condition had 16 participants (2 females) and the BRIGHT condition also had 16 participants (5 females). Participants’ ages ranged from 18 to 39 ($M = 27.31$). According to the post-experiment questionnaire, 56% of our participants had no prior VR experience. The distance judgment results are shown in Figure 3. Participants have a similar perception of distance in two different conditions (BRIGHT and DARK). On average, participants with the bright condition walked 87.87% of the actual target distance, and participants with the dark condition walked 89.25% of the actual target distance. Shapiro–Wilk tests for 2, 3, 4 and 5 meters each found that the data are normally distributed ($p > 0.05$). An ANOVA test showed no significant change in response to the bright and dark condition $F(1, 30) = 0.060, p = 0.808$. As expected, it also showed that distance judgments significantly changed in response to target distance $F(3, 90) = 850.74, p < 0.001$. There was no interaction between display condition and target distance $F(3, 90) = 0.268$ with $p = 0.848$. Since ANOVA showed no difference, we also ran equivalence test to see if two groups might be the same with $\alpha = 0.05$ and equivalence bounds ($\Delta_L = -5.0, \Delta_U = 5.0$). The test indicated that the two conditions were not statistically equivalent ($t(30) = 0.902, p = 0.187$).

One participant in the post-experiment questionnaire mentioned that the tracking system didn’t work well. This participant was not included in our analysis. We identified and resolved the problem before the next participant. This demonstrates that it is useful to

explicitly ask participants about any issues they might see in VR so that the experimenter can quickly address the issue. Participants were instructed not to count their steps. However, one participant reported that he counted his steps after completing the first experimental trial. This trial was excluded from the analysis and we reminded the participant to not count steps for the remaining trials.

3 DISCUSSION

Since we didn't artificially brighten the scene in our BRIGHT condition, it is effectively a baseline that we can compare to other studies with similar experimental designs to confirm that our results are consistent. In particular, Creem-Regehr et al. [2015] found very similar results with an Oculus DK2 and 3–6m distances. Li et al. [2016] reports participants walked 89.6% with a DK2 for distances 2–5m. Finally, Buck et al. [2018] found participants walked 83% of the way to targets with an Oculus CV1 over longer distances of 5–10m. In general, our results appear consistent with other studies.

Our results suggest that large brightness manipulations do not significantly impact absolute, egocentric distance judgments. In contrast, other studies varying periphery brightnesses show that brightness manipulations do influence distance judgments [Jones et al. 2012; Li et al. 2018] as measured by direct blind-walking. These results are challenging to reconcile because both studies effectively manipulate the brightness of the periphery. This discrepancy might be explained by the edge surrounding the 3D scene (in Li et al. [2018]), the texture of the periphery, or the contrast between the periphery and the scene. More research is necessary to determine why our experiment produced different results.

Similarly, previous work involving more subjective measures and verbal reports indicates that bright surfaces appear more distant than dark [Matusiak 2004; Oberfeld et al. 2010; Stamps III 2007; von Castell et al. 2018]. Although the surfaces were much darker in our DARK condition, we did not see a similar result. One possible explanation for this difference may be related to previous work showing that the quality of graphics influences verbal reports but not action-based measures [Kunz et al. 2009]. It is possible that brightness in our experiment would have effected verbal reports but did not influence direct blind-walking.

When we reduced the brightness in the DARK condition, we also effectively reduced the contrast between the dark and bright items in the scene. We chose this manipulation because it is most consistent with a brightness adjustment that might be built into HMDs and consistent with 3D scenes with dark environments. Contrast is naturally reduced in foggy environments and at long distances due to aerial perspective cues. In reduced cue real environments, studies suggest that reducing contrast may increase perceived distance [Farnè 1977; O'Shea et al. 1994]. Since our study effectively reduced brightness and contrast in the DARK condition, it is possible that darkening the scene and reducing contrast influences distance perception in opposite ways and cancel each other out.

Although specifications like the field of view are commonly reported, HMD manufacturers and research paper authors often do not publish HMD brightnesses. It is possible that over the past two dozen years, there has been a trend of increasing brightness in HMDs as display technology improves. Given the lack of detailed information, our work may increase confidence in comparing direct

blind-walking research papers which may use different HMDs with unknown differences in brightness.

4 LIMITATIONS & FUTURE WORK

Although our work shows that overall brightness does not influence distance judgments, it has limitations. For example, it uses only one response measure (blind walking), one indoor environment, two brightnesses, and a limited range of distances. There are four areas of future work help clarify how brightness influences distance judgments in HMDs. First, since previous work has found that the quality of graphics influences verbal reports but not blind walking [Kunz et al. 2009], it may be worthwhile to determine if a similar effect occurs when brightness is varied. This could help explain why other studies that vary paint colors find brightness changes spatial judgments [Matusiak 2004; Oberfeld et al. 2010; Stamps III 2007; von Castell et al. 2018]. Second, since previous work on overall brightness does not utilize direct blind walking, our experiment could be reproduced in the real world by having participants wear sunglasses or in rooms with different paint colors. Third, more variations of the Li et al. [2018] study could try various combinations of periphery brightness and 3D scene brightness. This work might provide additional insights that might help explain why periphery brightness influences distance judgments but varying overall brightness does not. Fourth, the current work could be extended to include a bright condition with reduced contrast. This additional condition would provide more detailed insights into how contrast and brightness might independently influence distance judgments.

Our work is limited to head-mounted displays. However, brightness may or may not play a role in distance judgments in other contexts. For example, with large screen displays, the overall brightness of the screen, and the overall brightness of the room could be factors. Changing the brightness of the room may be more analogous to previous work, which changes the brightness of the periphery in HMDs.

5 CONCLUSION

We performed a study comparing HMD distance judgment performance in a bright and artificially darkened virtual environment. Our results show that distance underestimation is not affected by the overall brightness of the scene. Since HMD screen brightness is often not reported by researchers or manufacturers, it is useful to know that we can compare two direct blind-walking studies that use HMDs with different brightnesses. People may wish to use dark VR environments for a variety of useful reasons. Manufacturers may wish to reduce energy usage, users may wish to relieve eye-strain by reducing brightness, and training facilities may wish to decrease brightness to simulate dark environments. Our results indicate that, in all of these cases, distance judgments as indicated by direct blind-walking are not negatively influenced. Although more work is needed to address the limitations of this study, our results suggest that virtual scenes can be made very dark without impairing the user's ability to judge distances.

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